THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 9 Solution
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1. (a) Let S and N be two random variables. Define the *conditional variance* of S given N by

$$\mathrm{Var}(S|N) = \mathbb{E}\left[(S - \mathbb{E}[S|N])^2|N\right],$$

which is a function of N. Assume that $\mathrm{Var}(S) < \infty$. Prove the following *law of total variance*:

$$\mathrm{Var}(S) = \mathbb{E}[\mathrm{Var}(S|N)] + \mathrm{Var}(\mathbb{E}[S|N]).$$

(b) Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and variance σ^2, N be an independent nonnegative integer-valued random variable and $S = \sum_{i=1}^{N} X_i$. Find $\mathrm{Var}(S)$ in terms of $\mathbb{E}[N]$ and $\mathrm{Var}(N)$.

Solution:

(a) We start from a more convenient expression of $\mathrm{Var}(S|N)$:

$$\begin{aligned}
\mathrm{Var}(S|N) &= \mathbb{E}\left[(S - \mathbb{E}[S|N])^2|N\right] \\
&= \mathbb{E}\left[S^2|N\right] - \mathbb{E}\left[2S \cdot \mathbb{E}[S|N] \mid N\right] + \mathbb{E}\left[(\mathbb{E}[S|N])^2|N\right] \\
&= \mathbb{E}[S^2|N] - 2\mathbb{E}[S|N]\mathbb{E}[S|N] + (\mathbb{E}[S|N])^2 \\
&= \mathbb{E}[S^2|N] - (\mathbb{E}[S|N])^2,
\end{aligned}$$

where the third equality is due to the fact that $\mathbb{E}[S|N]$ is a function of N and thus can be taken out of the conditional expectation (which is taken with respect to S). It follows that

$$\begin{aligned}
\mathbb{E}[\mathrm{Var}(S|N)] + \mathrm{Var}(\mathbb{E}[S|N]) &= \mathbb{E}[\mathbb{E}[S^2|N] - (\mathbb{E}[S|N])^2] + \mathrm{Var}(\mathbb{E}[S|N]) \\
&= \mathbb{E}[S^2] - \mathbb{E}[(\mathbb{E}[S|N])^2] + \left(\mathbb{E}[(\mathbb{E}[S|N])^2] - (\mathbb{E}[\mathbb{E}[S|N]])^2\right) \\
&= \mathbb{E}[S^2] - (\mathbb{E}[S])^2 = \mathrm{Var}(S).
\end{aligned}$$

(b) We leverage the result in part (a). Note that

$$\begin{aligned}
\mathbb{E}[S|N = n] &= \mathbb{E}\left[\sum_{i=1}^{N} X_i \mid N = n\right] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = n\mu, \\
\mathrm{Var}(S|N = n) &= \mathrm{Var}\left(\sum_{i=1}^{N} X_i \mid N = n\right) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = n\sigma^2.
\end{aligned}$$

Thus $\mathbb{E}[\mathrm{Var}(S|N)] = \mathbb{E}[\sigma^2 N] = \sigma^2\mathbb{E}[N]$ and $\mathrm{Var}(\mathbb{E}[S|N]) = \mathrm{Var}(\mu N) = \mu^2\mathrm{Var}(N)$. Hence by part (a), we have

$$\mathrm{Var}(S) = \mathbb{E}[\mathrm{Var}(S|N)] + \mathrm{Var}(\mathbb{E}[S|N]) = \sigma^2\mathbb{E}[N] + \mu^2(\mathrm{Var}(N)).$$

2. Let $X_0 = x$ and X_i $(i \geq 1)$ be independent uniform (0,1) random variables. Define N by

$$N = \min\{n : X_n < X_{n-1}\}$$

Let $f(x) = \mathbb{E}[N]$.

(a) Derive an integral equation for $f(x)$ by conditioning on X_1.

(b) Differentiate both sides of the equation derived in part (a) to obtain a differential equation for f.

(c) Solve the resulting equation obtained in part (b).

(d) For a second approach to determining $f(x)$, argue that

$$P(N > k) = \frac{(1-x)^k}{k!}$$

(e) Use part (d) to obtain $f(x)$.

Solution:

(a) Note that

$$f(x) = \mathbb{E}[N] = \int_0^1 \mathbb{E}[N|X_1 = y] f_{X_1}(y) dy = \int_0^1 \mathbb{E}[N|X_1 = y] dy.$$

For $\mathbb{E}[N|X_1 = y]$, if $y < x$ then $N = 1$; if $y \geq x$, by the independence of X_i, we have $N = 1 + N'$ where N' is defined similarly to N but with $X_0 = y$. Thus

$$\mathbb{E}[N|X_1 = y] = \begin{cases} 1, & y < x, \\ 1 + f(y), & y \geq x. \end{cases}$$

Hence, $f(x) = 1 + \int_x^1 f(y) dy$.

(b) By Newton-Leibniz formula, we have $f'(x) = -f(x)$.

(c) Note that $f(1) = 1$ since $N = 1$ if $X_1 < X_0$. Thus we have $f(x) = e^{1-x}$.

(d) We have

$$\begin{aligned} P(N > n) &= P(x < X_1 < X_2 < \cdots < X_n) \\ &= P(\min\{X_1, X_2, \ldots, X_n\} > x,\ X_1 < X_2 < \cdots < X_n). \end{aligned}$$

Note that $P(\min\{X_1, X_2, \ldots, X_n\} > x) = (1-x)^n$. Since X_i are i.i.d., by symmetry, we have $P(X_1 < X_2 < \cdots < X_n) = \frac{1}{n!}$ (since their are $n!$ possible orderings), and the event $\{X_1 < X_2 < \cdots < X_n\}$ is independent of the event $\{\min\{X_1, X_2, \ldots, X_n\} > x\}$. We thus conclude that

$$P(N > n) = P(\min\{X_1, X_2, \ldots, X_n\} > x) P(X_1 < X_2 < \cdots < X_n) = \frac{(1-x)^n}{n!}.$$

(e) We first prove the following general and useful result: for any nonnegative integer-valued random variable N, we have

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} P\{N > n\}.$$

Indeed, starting from the right-hand side, we have

$$\begin{aligned}\sum_{n=0}^{\infty} P\{N > n\} &= \sum_{n=0}^{\infty} \sum_{k=n+1}^{\infty} P(N = k) \\ &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} P(N = k) \\ &= \sum_{k=1}^{\infty} kP(N = k) = \mathbb{E}[N].\end{aligned}$$

Here, the infinite sum is interchangable since $P(N = k) \geq 0$ for all k.
Now by part (d), we have

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} P\{N > n\} = \sum_{n} \frac{(1-x)^n}{n!} = e^{1-x}.$$

3. There are n types of coupons with each type labeled by $1, 2, \ldots, n$, respectively. The type of each new coupon obtained is independent of past selections and is equally likely to be any of the n types.

   (a) Suppose that one continuously collects the coupons until he obtains at least one of coupon with type $1, \ldots, k$, respectively, where $1 \leq k \leq n$. Argue that the probability that the last coupon collected is of type i for $1 \leq i \leq k$ is $\frac{1}{k}$.

   (b) Now suppose that one continuously collects the coupons until a complete set of at least one of each type is obtained. Find the probability that there is exactly one type-i coupon in the final collection.

   (c) Under the same conditions in (b), find the expected number of types that appear exactly once in the final collection.

   Hint: Let T be the random variable representing the number of types of coupons that appear before the first appearance of type-i. Utilize the trick that each possible collection happens equally likely to its reverse order.

   Solution:

   (a) This is obvious since the last coupon collected must be one of the k types, and by symmetry, each type has equal probability to be the last one.

   (b) Let S_i be the boolean random variable of the event "type i coupon only appears once in the final set" and T be the random variable representing the number of type of coupons appear before type i. By conditioning on T, we have

$$P(S_i = 1) = \sum_{j=0}^{n-1} P(S_i = 1 \ \cap \ T = j \ ) = \sum_{j=0}^{n-1} P(S_i = 1 | T = j) P(T = j).$$

   We first claim that $P(T = j) = 1/n$ for any $0 \leq j \leq n-1$. This follows from a similar argument to part (a): each coupon type has equal probability to be the i-th one to first appear, and thus $P(T = j) = 1/n$ for any j.

Now we consider $P(S_i = 1|T = 0)$. This means type i appears in the first draw and never appear until all types are collected. For each sequence of draws satisfying this condition, the order-reversed sequence of draws happens equally likely, which is characterized by that type i is the last type to appear. By part (a), this happens with probability $1/n$. Thus $P(S_i = 1|T = 0) = 1/n$.

For $P(S_i = 1|T = j)$ where $j > 0$, this conditional event means that type i never appears until we finish collected the remaining $n - j - 1$ types of coupon. By reverting the order of draws after the draw of type i, we see that the probability of this event is the same as the probability that type i appears last among the remaining $n - j$ types. Thus $P(S_i = 1|T = j) = 1/(n - j)$ for any $j \geq 1$.

Therefore,

$$P(S_i = 1) = \frac{1}{n}\sum_{j=0}^{n-1} P(S_i = 1|T = j) = \frac{1}{n}\sum_{j=0}^{n-1}\frac{1}{n-j} = \frac{1}{n}\sum_{k=1}^{n}\frac{1}{k}.$$

(c) The expected number of types appear exactly once is given by

$$\sum_{i=1}^{n}\mathbb{E}(S_i) = \sum_{i=1}^{n} P(S_i = 1) = \sum_{i=1}^{n}\frac{1}{n}\sum_{k=1}^{n}\frac{1}{k} = \sum_{k=1}^{n}\frac{1}{k}.$$